Schrödinger follmer sampler

Schrödinger Bridge问题（SBP）

考虑在 $t = 0$ 时刻观察到粒子分布为 $\tilde{\nu}$，在 $t = 1$ 时刻观察到粒子分布为 $\tilde{\mu}$

考虑标准布朗运动作为先验与参照，有先验测度 $\mathbf{P} = \int \mathbf{W}_x dx$，$\mathbf{W}_x$ 是从 x 点出发的维纳测度, 在任何时刻 t，它的边际分布都是勒贝格测度 $\mathscr{L}$.

考虑在路径空间 $\Omega = C([0,1], \mathbb{R}^p)$ 中，每一个元素 ω 则是一条连续的路径，有投影算子Z_t满足

$Z_t(\omega) = \omega_t$

即给出路径ω在t时刻的位置。

如果 $\tilde{\nu}$ 到 $\tilde{\mu}$ 的演化不符合单纯的布朗运动规律（非自然扩散），那么在这两个分布之间，粒子最可能的路径概率分布 $\mathbf{Q}^*$ 是什么样的?

约束条件 $\mathbf{Q}_0 = \tilde{\nu}, \mathbf{Q}_1 = \tilde{\mu}$下，有路径的概率测度 $\mathbf{Q}^* \in \mathcal{P}(\Omega)$, 并且t时刻的边际分布满足 $Q_t^* = (Z_t)_\# \mathbf{Q}^* = \mathbf{Q}^* \circ Z_t^{-1}$，$t \in [0,1]$(其实是用原像定义t时刻粒子落在某个区域的概率：$\mathbf{Q}_t^*(A) = \mathbf{Q}^*(Z_t^{-1}(A))$)

$$Q^* \in \arg\min \mathbb{D}_{\mathrm{KL}}(\mathbf{Q} \| \mathbf{P}),$$

其中

$$\mathbb{D}_{\mathrm{KL}}(\mathbf{Q} \| \mathbf{P}) = \int \log\left(\frac{d\mathbf{Q}}{d\mathbf{P}}\right) d\mathbf{Q}$$

根据Radon-Nikodym 定理，只有当 $\mathbf{Q} \ll \mathbf{P}$ 时，dQ/dP 才在 $\mathbf{P}$-几乎处处存在且是可测的。

SBP在上述边界约束条件下，寻找一个与布朗运动在KL散度上最接近的路径概率测度，目标是$\mathbf{Q}^*$ 和 $\mathbf{P}$ 必须在同一个路径空间簇内，相当于在先验的基础上做修正，尽可能的模仿标准布朗运动

schrodinger follmer diffusion SDE

schrodinger follmer diffusion将t=0时的初始分布δ_0传输到t=1时的目标分布μ。

$$f(x) = \frac{d\mu}{dN(0, \mathbf{I}_p)}(x), x \in \mathbb{R}^p.$$

设热半群Q_t:

$$Q_t f(x) = \mathbb{E}_{Z \sim N(0, \mathbf{I}_p)}\left[f(x + \sqrt{t}Z)\right], \quad t \in [0,1].$$

Schr¨odinger-F¨ollmer diffusion process 定义为：

$$dX_t = -\nabla_x U(X_t, t) dt + dB_t, X_0 = 0, t \in [0,1]$$

U是势能函数：

$$U(x,t) = -\log Q_{1-t}f(x).$$

随机过程$\{X_t\}$是上面SDE的解，满足初始测度$\tilde{\nu} = \delta_0$(dirac测度)，$\tilde{\mu} = \mu$

SDE的漂移项可以写为：

$$b(x,t) \equiv -\nabla_x U(x,t) = \frac{\mathbb{E}_Z[\nabla f(x + \sqrt{1-t}Z)]}{\mathbb{E}_Z[f(x + \sqrt{1-t}Z)]}, \quad x \in \mathbb{R}^p, t \in [0,1],$$

C1：

drift term b满足线性增长条件

$$\|b(x,t)\|_2^2 \le C_0(1 + \|x\|_2^2), x \in \mathbb{R}^p, t \in [0,1]$$

C2:

drift term b满足lipschitz条件：

$$\|b(x,t) - b(y,t)\|_2 \le C_1\|x-y\|_2, \quad x, y \in \mathbb{R}^p, t \in [0,1],$$

Proposition 2.1. *Assume (C1) and (C2) hold, then the Schrödinger-Föllmer SDE (2.6) has a unique strong solution* $\{X_t\}_{t\in[0,1]}$ *with* $X_0 \sim \delta_0$ *and* $X_1 \sim \mu$.

Proposition 2.1 说明当 C1和C2成立， 上述边值SDE有唯一strong solution。

接下来只需要对C1和C2满足的sde采样就能得到想要的目标分布：

考虑Euler-Maruyama离散化

We use the Euler-Maruyama discretization for the SDE (2.6) with a fixed step size. Let

$$t_k = k \cdot s, \ k = 0, 1, \ldots, K, \text{ with } s = 1/K,$$

and set $Y_{t_0} = 0$. Then the Euler-Maruyama discretization of (2.6) has the form

$$Y_{t_{k+1}} = Y_{t_k} + sb(Y_{t_k}, t_k) + \sqrt{s}\,\epsilon_{k+1}, \ k = 0, 1, \ldots, K-1, \tag{2.16}$$

其中，drift term定义为

$$b(Y_{t_k}, t_k) = \frac{\mathbb{E}_Z[\nabla f(Y_{t_k} + \sqrt{1-t_k}Z)]}{\mathbb{E}_Z[f(Y_{t_k} + \sqrt{1-t_k}Z)]}, Z \sim N(0, \mathbf{I}_p).$$

实际运算中分子分母中的期望 需要对其进行蒙特卡洛估计， 这需要$f, \nabla f$是可以计算的：

目标分布可以写成 归一化的形式：

$$\mu(x) = \frac{1}{C}\exp(-V(x)), x \in \mathbb{R}^p$$

Radon-Nikodym导数 f 可以写为

> where V has a known form, but C may be unknown. The Radon-Nikodym derivative of μ with respect to $N(0, \mathbf{I}_p)$ can be written as $f(x) = C^{-1}(2\pi)^{p/2}g(x)$, where
>
> $$g(x) = \exp\left(-V(x) + \frac{1}{2}\|x\|^2\right), \quad x \in \mathbb{R}^p. \tag{2.10}$$

于是漂移项可以改写为显式进行蒙特卡洛模拟计算的形式：

$$\mathbb{E}_Z[\nabla f(Y_{t_k} + \sqrt{1-t_k}Z)] = \frac{1}{\sqrt{1-t_k}}\mathbb{E}_Z[Zf(Y_{t_k} + \sqrt{1-t_k}Z)].$$

This identity enables us to avoid the calculation of ∇f. The drift term can be rewritten as

$$b(Y_{t_k}, t_k) = \frac{\mathbb{E}_Z[Zf(Y_{t_k} + \sqrt{1-t_k}Z)]}{\mathbb{E}_Z[f(Y_{t_k} + \sqrt{1-t_k}Z)] \cdot \sqrt{1-t_k}}.$$

Since g in (2.10) is proportional to f up to a multiplicative constant independent of x, we can express b in terms of g as

$$b(Y_{t_k}, t_k) = \frac{\mathbb{E}_Z[\nabla g(Y_{t_k} + \sqrt{1-t_k}Z)]}{\mathbb{E}_Z[g(Y_{t_k} + \sqrt{1-t_k}Z)]}, \tag{2.18}$$

or

$$b(Y_{t_k}, t_k) = \frac{\mathbb{E}_Z[Zg(Y_{t_k} + \sqrt{1-t_k}Z)]}{\mathbb{E}_Z[g(Y_{t_k} + \sqrt{1-t_k}Z)] \cdot \sqrt{1-t_k}}. \tag{2.19}$$

$$\tilde{b}_m(Y_{t_k}, t_k) = \frac{\frac{1}{m}\sum_{j=1}^m \left[\nabla g\left(Y_{t_k} + \sqrt{1-t_k}Z_j\right)\right]}{\frac{1}{m}\sum_{j=1}^m \left[g\left(Y_{t_k} + \sqrt{1-t_k}Z_j\right)\right]}, \quad k = 0, \ldots, K-1,$$

或

$$\tilde{b}_m(Y_{t_k}, t_k) = \frac{\frac{1}{m}\sum_{j=1}^m \left[Z_j g\left(Y_{t_k} + \sqrt{1-t_k}Z_j\right)\right]}{\frac{1}{m}\sum_{j=1}^m \left[g\left(Y_{t_k} + \sqrt{1-t_k}Z_j\right)\right] \cdot \sqrt{1-t_k}}, \quad k = 0, \ldots, K-1.$$

于是就可以用如下方法采样：

$$\tilde{Y}_{t_{k+1}} = \tilde{Y}_{t_k} + s\tilde{b}_m(\tilde{Y}_{t_k}, t_k) + \sqrt{s}\epsilon_{k+1}, k = 0, 1, \ldots, K-1,$$

伪代码如下：

Algorithm 2 SFS for $\mu = \exp(-V(x))/C$ with Monte Carlo estimation of the drift term

1: Input: $V(x)$, m, K. Initialize $s = 1/K$, $\widetilde{Y}_{t_0} = 0$.
2: **for** $k = 0, 1, \ldots, K-1$ **do**
3: Sample $\epsilon_k \sim N(0, \mathbf{I}_p)$,
4: Compute $\tilde{b}_m$ according to (2.20) or (2.21),
5: Update $\widetilde{Y}_{t_{k+1}} = \widetilde{Y}_{t_k} + s\tilde{b}_m\left(\widetilde{Y}_{t_k}, t_k\right) + \sqrt{s}\,\epsilon_{k+1}$.
6: **end for**
7: Output: $\{\widetilde{Y}_{t_k}\}_{k=1}^K$

> **Tip**
>
> **热半群：**是热方程的解算子：$u(x,t) = (P_t f)(x)$
>
> 可以表示为卷积， 满足半群的性质；扩散模型中，前向扩散过程（Forward Process）本质上就是通过热半群（或其变体，如 Ornstein-Uhlenbeck 半群）将复杂的原始数据分布逐渐转化为简单的正态分布。

理论性能保证

C3假设：关于x lipschitz连续， 关于t holder连续

$$\|b(x,t) - b(y,s)\|_2 \le C_1\left(\|x-y\|_2 + |t-s|^{1/2}\right), x, y \in \mathbb{R}^p \text{ and } t, s \in [0,1],$$

定义衡量分布间距的W_d度量：

$$W_d(\nu_1, \nu_2) = \inf_{\nu \in \mathcal{D}(\nu_1, \nu_2)} \left(\iint_{\mathbb{R}^p \mathbb{R}^p} \|\theta_1 - \theta_2\|_2^d \, d\nu(\theta_1, \theta_2)\right)^{1/d}.$$

这里只介绍不能直接计算采样只能monte carlo的算法2的性能保证

C4假设： 势能U的强凸性假设

$$U(x,t) - U(y,t) - \nabla U(y,t)^{\mathrm{T}}(x-y) \ge (M/2)\|x-y\|_2^2$$

并且$M < C_1$ given in C3

下面给出性能保证：

Theorem 4.2. *Assume (C1), (C3) and (C4) hold, g and ∇g are Lipschitz continuous, and g is bounded below away from 0. Then, for $s < \frac{2}{C_1+M} < 1$,*

$$W_2(Law(\widetilde{Y}_{t_K}), \mu) \le \mathcal{O}(\sqrt{ps}) + \mathcal{O}\left(\sqrt{\frac{p}{\log(m)}}\right),$$

where m is the size of the normal random sample used in approximating the drift term b in (2.20) or (2.21).

进一步可以有：

Theorem 4.3. *Assume that, in addition to the conditions of Theorem 4.2, g is bounded above. Then, for $s < \frac{2}{C_1+M} < 1$,*

$$W_2(Law(\widetilde{Y}_{t_K}), \mu) \leq \mathcal{O}(\sqrt{ps}) + \mathcal{O}\left(\sqrt{\frac{p}{m}}\right).$$

证明细节（th4.2-4.3）：

需要的引理:

Lemma A.3. *(Lemma 1 in Dalalyan (2017a) and Lemma 2 in Dalalyan and Karagulyan (2019)). Denote $\Delta_k = X_{t_k} - \widetilde{Y}_{t_k}$ and $\omega_k = b(\widetilde{Y}_{t_k}, t_k) - b(X_{t_k}, t_k)$ with $k = 0, 1, \ldots, K$. Assume conditions (C2) and (C4) hold, and $s < 2/(C_1 + M) < 1$, then*

$$\|\Delta_k - s\omega_k\|_2 \leq \rho\|\Delta_k\|_2,$$

where $\rho = 1 - sM \in (0, 1)$.

Lemma A.4. *If g and ∇g are Lipschitz continuous, and g has the lower bound greater than 0, then for any $R > 0$,*

$$\sup_{\|x\|_2 \leq R, t\in[0,1]} \mathbb{E}\left[\|b(x,t) - \tilde{b}_m(x,t)\|_2^2\right] \leq \mathcal{O}\left(\frac{p\exp(R^2)}{m}\right).$$

Moreover, if g has a finite upper bound, then

$$\sup_{x\in\mathbb{R}^p, t\in[0,1]} \mathbb{E}\left[\|b(x,t) - \tilde{b}_m(t,x)\|_2^2\right] \leq \mathcal{O}\left(\frac{p}{m}\right).$$

Lemma A.5. *Assume that g is γ-Lipschitz continuous and has the lower bound greater than 0, that is, $g \geq \xi > 0$ for a positive and finite constant ξ, then for $k = 0, 1, \ldots, K$,*

$$\mathbb{E}\|\widetilde{Y}_{t_k}\|_2^2 \leq \frac{6\gamma^2}{\xi^2} + 3p.$$

Lemma A.6. *If g and ∇g are Lipschitz continuous, and g has the lower bound greater than 0, then for $k = 0, 1, \ldots, K$ and $t \in [0,1]$,*

$$\mathbb{E}\left\|b(\widetilde{Y}_{t_k}, t_k) - \tilde{b}_m(\widetilde{Y}_{t_k}, t_k)\right\|_2^2 \leq \mathcal{O}\left(\frac{p}{\log(m)}\right),$$

$$\mathbb{E}\left\|b(X_t, t) - \tilde{b}_m(X_t, t)\right\|_2^2 \leq \mathcal{O}\left(\frac{p}{\log(m)}\right).$$

Moreover, if g has a finite upper bound, then

$$\mathbb{E}\left\|b(\widetilde{Y}_{t_k}, t_k) - \tilde{b}_m(\widetilde{Y}_{t_k}, t_k)\right\|_2^2 \leq \mathcal{O}\left(\frac{p}{m}\right),$$

$$\mathbb{E}\left\|b(X_t, t) - \tilde{b}_m(X_t, t)\right\|_2^2 \leq \mathcal{O}\left(\frac{p}{m}\right).$$

th4.2的证明：

目标是bound住误差在K时的L2范数的上界

$$\Delta_k = X_{t_k} - \widetilde{Y}_{t_k}.$$

先拆开成误差传播的形式

$$\begin{aligned}\Delta_{k+1} &= \Delta_k + (X_{t_{k+1}} - X_{t_k}) - (\tilde{Y}_{t_{k+1}} - \tilde{Y}_{t_k}) \\ &= \Delta_k - s\left[\tilde{b}_m(\tilde{Y}_{t_k}, t_k) - \tilde{b}_m(\tilde{Y}_{t_k} + \Delta_k, t_k)\right] + \int_{t_k}^{t_{k+1}} \left[b(X_t, t) - \tilde{b}_m(X_{t_k}, t_k)\right] dt.\end{aligned}$$

借助lemma A.3放缩第一项

$$\begin{aligned}&\left\|\Delta_k - s\left[\tilde{b}_m(\tilde{Y}_{t_k}, t_k) - \tilde{b}_m(\tilde{Y}_{t_k} + \Delta_k, t_k)\right]\right\|_{L_2} \\ \leq &\left\|\Delta_k - s\left[b(\tilde{Y}_{t_k}, t_k) - b(\tilde{Y}_{t_k} + \Delta_k, t_k)\right]\right\|_{L_2} + s\left\|b(\tilde{Y}_{t_k}, t_k) - \tilde{b}_m(\tilde{Y}_{t_k}, t_k)\right\|_{L_2} \\ &+ s\left\|\tilde{b}_m(\tilde{Y}_{t_k} + \Delta_k, t_k) - b(\tilde{Y}_{t_k} + \Delta_k, t_k)\right\|_{L_2} \\ \leq &\ \rho\|\Delta_k\|_{L_2} + s\left\|b(\tilde{Y}_{t_k}, t_k) - \tilde{b}_m(\tilde{Y}_{t_k}, t_k)\right\|_{L_2} + s\left\|b(\tilde{Y}_{t_k} + \Delta_k, t_k) - \tilde{b}_m(\tilde{Y}_{t_k} + \Delta_k, t_k)\right\|_{L_2}.\end{aligned}$$

借助 lemma A.6 得到上界

$$\left\|\Delta_k - s\left[\tilde{b}_m(\mathbf{\tilde{Y}}_{t_k}, t_k) - \tilde{b}_m(\mathbf{\tilde{Y}}_{t_k + \Delta_k}, t_k)\right]\right\|_{L_2} \leq \rho\|\Delta_k\|_{L_2} + s \cdot \mathcal{O}\left(\sqrt{\frac{p}{\log(m)}}\right).$$

借助C3假设和 lemma A.2和A.6得到第二项的上界

$$\left\|\int_{t_k}^{t_{k+1}} (b(X_t, t) - \tilde{b}_m(X_{t_k}, t_k)) dt\right\|_{L_2} \leq \mathcal{O}(\sqrt{ps^{3/2}}) + \mathcal{O}\left(s\sqrt{\frac{p}{\log(m)}}\right).$$

进而有误差传播不等式

$$\|\Delta_{k+1}\|_{L_2} \le \rho\|\Delta_k\|_{L_2} + \mathcal{O}(\sqrt{ps^{3/2}}) + \mathcal{O}\left(s\sqrt{\frac{p}{\log(m)}}\right).$$

递推并假设t0时误差为0，得到上界估计：

$$\|\Delta_{k+1}\|_{L_2} \le \rho^{k+1}\|\Delta_0\|_{L_2} + \mathcal{O}(\sqrt{sp}) + \mathcal{O}\left(\sqrt{\frac{p}{\log(m)}}\right).$$

$$\begin{aligned} W_2(\mathrm{Law}(\tilde{Y}_{t_K}), \mu) &\le \rho^K\|\Delta_0\|_{L_2} + \mathcal{O}(\sqrt{sp}) + \mathcal{O}\left(\sqrt{\frac{p}{\log(m)}}\right) \\ &\le \mathcal{O}(\sqrt{sp}) + \mathcal{O}\left(\sqrt{\frac{p}{\log(m)}}\right). \end{aligned}$$

可见证明需要引理中对于漂移项的误差进行估计，